

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination Oct 2017

PH322 Phytochemistry- II

Date: 04.10.2017 , Wednesday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1 Attempt all questions. Each question is of one mark. 20
1. Beta carotene on distillation yielded ____ + ____ + ____
[A] Benzene+ m- cresol + naphthalene
[B] Toluene+ m- xylene + 2,6- dimethyl naphthalene
[C] Phenol +m - xylene + 2,6- dimethyl naphthalene
[D] Acetic Acid
 2. One of following is aldehyde
[A] Menthol
[B] Abietic Acid
[C] Citral
[D] Retenone
 3. How many ionone molecules are obtained from retinol ?
[A] 1
[B] 2
[C] 3
[D] 4
 4. Menthol is obtained from _____.
[A] Pipermint oil
[B] Jasmine flower oil
[C] Rose flower oil
[D] Turpentine
 5. _____ are made up of isoprene units.
[A] Isopropenoids
[B] Phytone
[C] Flvonoids
[D] Terpenoids
 6. Flavonoids are considered as one of _____ type of compounds.
[A] Acidic
[B] Phenyl alkyl amine
[C] Phenyl Isopropenoidal
[D] Phenyl butyl
 7. Volatile oils are soluble in _____.
[A] Acid
[B] Base
[C] Water
[D] Chloroform

8. Oxygenated carotenoid is termed as _____.
 [A] Xanthophylls
 [B] Carotenes
 [C] Lycopene
 [D] Vitamin A
9. One of the following is the commercial source of Oleonolic acid
 [A] Linseed oil
 [B] Turpentine oil
 [C] Citrus oil
 [D] Clove Oil
10. Beta-carotene is _____ type of terpenoid.
 [A] Sesquiterpenoid
 [B] Triterpenoid
 [C] Monoterpene
 [D] Tetraterpoid
11. Alpha- pinene on reaction with sulphuric acid yeilds _____.
 [A] alpha -terpeniol
 [B] beta-ionone
 [C] Pinic acid
 [D] Amino butyric acid
12. Molecular formula of isoprene is
 [A] C_5H_8
 [B] $C_{15}H_8$
 [C] C_5H_{18}
 [D] C_2H_5
13. Low molecular weight volatile terpenoids may be estimated from volatile oil using _____.
 [A] Gravimetry
 [B] Titrymetry
 [C] HPTLC
 [D] Gas Chromatography
14. _____ is responsible for color production.
 [A] Unsturation
 [B] Methylation
 [C] Oxydation
 [D] Reduction
15. _____ is not used as flavoring agent.
 [A] Raspberry
 [B] Rose oil
 [C] Jasmine oil
 [D] Vannila
16. Which method will be preferred to get premium quality rose oil?
 [A] Distillation
 [B] Percolation
 [C] Freactionation
 [D] Super Critical Fluid Extraction
17. _____ is used as natural color for clothes.
 [A] Caramel
 [B] Bixin
 [C] Indigo
 [D] Saffron
18. _____ is used as flavouring agent.
 [A] Bixin
 [B] Saffron
 [C] Caramine
 [D] Xanthophyll
19. Immediately, after spraying the perfume, we have smell of _____.
 [A] Top Note
 [B] Middle Note

- [C] Base Note
[D] All together

20. _____ can form complexes with flavonoids.

- [A] Aluminum
[B] Sodium
[C] Potassium
[D] Magnesium

SECTION – II

Q 2 Attempt any **TWO** of the following

- A Prove following: 05
1. Menthone is cyclic compound.
2. Menthone contains ketonic group.
- B Prove following: 05
1. One ring in alpha- pinene is four membered ring.
2. Limonene contains two double bond s, while only one of them inside ring.
- C Apply chemical approach to prove chemical structure of Geraniol. 05

Q 3 Attempt any **TWO** of the following

- A Write important chemical reactions involved in structure elucidation of aglycone of Apigenin. 05
- B Write a note on isolation, chemical structure and uses of lycopene. 05
- C Draw chemical structure of flavanyl ring and write other features of flavonoidal chemical structure. 05

SECTION – III

Q 4 Attempt any **FOUR** of the following

- A How will you select different chemicals to develop a perfume? 05
- B Classify Natural Colorants on the basis of chemical structure. 05
- C Write in brief about Betalains and Chlorophyll with special reference to their application as natural colorants. 05
- D Write botanical sources and chemicals of plants to be used as sweetening agents. 05
- E Write biological source, chemical constituent responsible of at least five natural products, used as source of volatile oil to formulate perfume. 05
- F Marine Natural Products are potential sources of novel drugs. Justify this sentence with examples. 05

Q 5 Attempt any **FOUR** of the following

- A Write the reactions to be performed to ensure functional nature of oxygen in given terpenoidal compound. 05
- B What does it mean by word Terpenoid. Classify terpenoids. 05
- C Write chemical reactions involved in elucidating the structure of Retinol. 05
- D Write important steps involved in biogenesis of beta-carotene in plants. 05
- E Do as directed, 05

- 1 Complete reaction
Acyclic Terpene ----->
Se

- 2 How will you determine C-Methyl group in terpenoidal structure.

- F Write botanical source and uses of Oleonic acid. Write important reactions involved in elucidating chemical structure of Oleonic acid. 05